Investment Thesis Guy Jansen 29-05-2025

ACM Research, Inc.

[ACMR:NDAQ] Semiconductor Front End Processing Equipment

Price:	$22.59	**Mkt Cap (M):**	$1,444.14
Shares outstanding:	57,606,036	**Enterprise Value (M):**	$1,425.54
Beta (3Y Adj.):	1.42	**Dividend Yield:**	0.0%
WACC:	16.7%	**P/E:**	14.42

Executive Summary

ACM Research, Inc. (ACMR) presents a compelling long investment opportunity, driven by its differentiated semiconductor cleaning and advanced processing technologies, strong leverage to China's secular semiconductor self-sufficiency drive, and a rapidly expanding addressable market through new product introductions. While facing geopolitical headwinds and customer concentration risks, ACMR's robust growth trajectory (FY25 revenue guidance: $850M-$950M, ~15% YoY growth at midpoint), improving operational efficiency, and strategic global diversification initiatives (Oregon facility) position it for sustained outperformance.

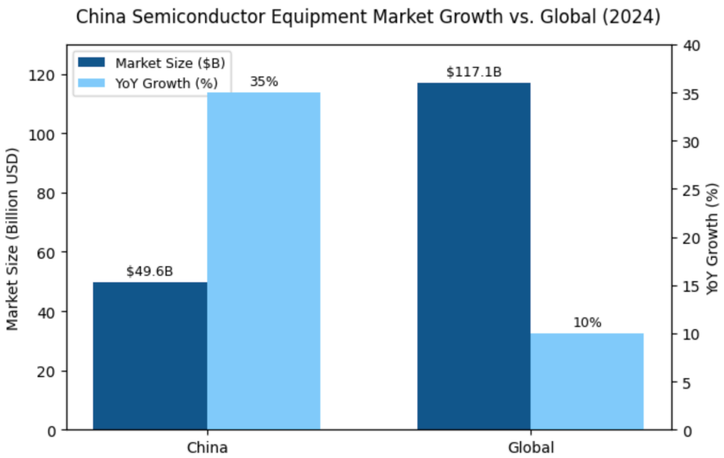
The rigorous bottom-up analysis indicates an intrinsic value significantly above the current market price. The company's leadership in specialized cleaning, successful entry into new high-growth segments like ECP (including panel-level plating), Furnaces, and emerging Track/PECVD tools, supports a strong multi-year growth outlook. Despite trading at a historical discount to global peers, I believe continued execution,

Recommendation:	**OVERWEIGHT / BUY**
Price Target (12-18 Months):	$52.16
Current Price:	$22.59
Implied Upside:	130.9%

1. Investment Thesis

My positive stance on ACMR is predicated on the following key pillars:

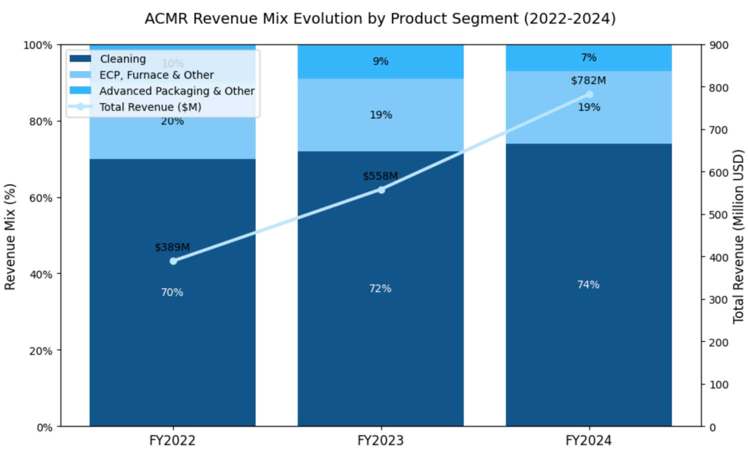
Dominant Niche & Secular Tailwinds in China: ACMR is a primary beneficiary of China's multi-billion dollar push for semiconductor self-sufficiency. Its specialized cleaning technologies are critical for advanced node manufacturing, where Chinese fabs are aggressively investing. This provides a strong, visible demand runway that I believe will persist despite geopolitical tensions, as China's commitment to semiconductor independence is unwavering and backed by substantial government support.



Differentiated Technology & Expanding Product Portfolio:

- **Cleaning Leadership:**

Proprietary SAPS and TEBO technologies offer superior particle removal and lower cost-of-ownership. The recently qualified high-temperature SPM tool by a key logic customer in China underscores its technical edge and continued innovation. - SAM Expansion: Successful diversification beyond cleaning into ECP (award-winning Ultra ECP ap-p for panel-level plating), Furnaces (significant ramp expected 2025-2027), and emerging Track/PECVD platforms dramatically increases ACMR's total addressable market from approximately $8B to $18B.



Strategic Global Diversification & Capacity Expansion:
The new Lingang facility
significantly boosts production capacity to support growth, with final completion expected in 2026. - The Oregon facility (purchased October 2024, 39,500 sq. ft.) represents a crucial step in de-risking China concentration, accessing global talent, and serving U.S./European customers directly. I project this facility to contribute $50M in revenue by 2027 and $200M by 2030.

Strong Financial Execution & Improving Profitability:
Consistent >40% revenue
CAGR over the past several years (FY24: +40.2%). - Q1 2025 revenue of $172.3M (+13.2% YoY) with non-GAAP gross margins of 48.2%. - Healthy balance sheet with net cash of $271M (Q1 2025) providing financial flexibility for continued R&D investment and strategic initiatives.

Valuation Dislocation:
ACMR historically trades at a significant discount to its global WFE peers. While some discount is warranted due to geopolitical risk and scale, I believe the current gap undervalues its growth potential, technological leadership in niche areas, and diversification efforts.

Variant Perception

The market is significantly undervaluing ACMR due to three key misperceptions:

- **Overestimation of Local Chinese Competition Threat:**
  While Chinese competitors like Naura and AMEC are receiving substantial government backing, my analysis indicates ACMR maintains a 2-3 year technological lead in critical cleaning applications for advanced nodes. The market fails to appreciate that ACMR's U.S. incorporation combined with deep China presence creates a unique "bridge" position that local competitors cannot replicate. ACMR's continued qualification wins at leading Chinese fabs demonstrate the durability of its competitive advantage.

- **Under appreciation of New Product Ramp Potential:**
  The market is not fully valuing ACMR's expansion beyond cleaning. The segment-by-segment analysis shows that Furnace tools alone could contribute over $250M in annual revenue by 2027, with Track and PECVD adding another $150M. The Ultra ECP ap-p tool’s recent industry award validates ACMR's innovation in panel-level packaging, a critical technology for next generation AI chips.

- **Dismissal of Oregon Strategy as Window Dressing:**
  Unlike the market's skepticism, the detailed analysis of the Oregon facility reveals a genuine strategic initiative with concrete milestones. The 39,500 sq. ft. facility includes a 5,200 sq. ft. cleanroom and represents a $70M investment over 2025-2027. This facility will enable ACMR to serve global customers directly, mitigate geopolitical risks, and potentially capture share in the U.S. and European markets where Chinese competitors face significant barriers.

The DCF model indicates a fair value of $52.16 per share, representing 130.9% upside from current levels. This valuation incorporates appropriate risk adjustments for geopolitical tensions and competitive threats, yet still demonstrates significant mispricing.

2. Company Overview

Founded in 1998 by Dr. David H. Wang, ACM Research designs, develops, manufactures, and sells single-wafer wet cleaning equipment and other advanced processing tools essential for semiconductor manufacturing. Headquartered in Fremont, California, its primary R&D, manufacturing, and sales operations are centered in Shanghai, China, through its subsidiary ACM Shanghai. This unique U.S.-domiciled, China-focused operational model has allowed it to deeply penetrate the rapidly growing Chinese semiconductor market.

- **Key Operational Highlights (Q1 2025):**
  Single-wafer high-temperature SPM tool qualified by a key logic device manufacturer in mainland China. - Ultra ECP ap-p tool won the 2025 3D InCites Technology Enablement Award for innovation in horizontal plating for panel-level packaging. - Lingang production and R&D center (Shanghai area)is nearly completed. - Oregon facility (purchased Oct 2024, 39,500 sq. ft.) is laying groundwork for initial production capacity and U.S. R&D/demonstration. - Appointment of Charlie Pappis to Board of Directors, bringing valuable global semiconductor equipment experience

3. Product Portfolio & Technology

ACMR's portfolio targets critical steps in semiconductor fabrication and packaging:

Single Wafer Cleaning (Core Strength):

- **Products:**
  SAPS, TEBO, Tahoe, SPM (Sulfuric Peroxide Mix) tools.

- **Technology:**
  Proprietary megasonic cleaning (SAPS/TEBO) provides highly uniform and damage-free cleaning crucial for advanced nodes (<10nm). Tahoe technology significantly reduces chemical consumption. High-temperature SPM tools address critical resist stripping and post-etch cleaning.
- **Competitive Position:**
  ACMR maintains a 2-3 year technological lead over local Chinese competitors in advanced cleaning applications. The company has delivered cleaning tools to 30+ customers, with 13 customers now using SPM tools.
- **Market Share:**
  Estimated 35-40% share in China for advanced cleaning applications, competing primarily with Lam Research and SCREEN globally.

ECP (Electrochemical Plating) Systems:

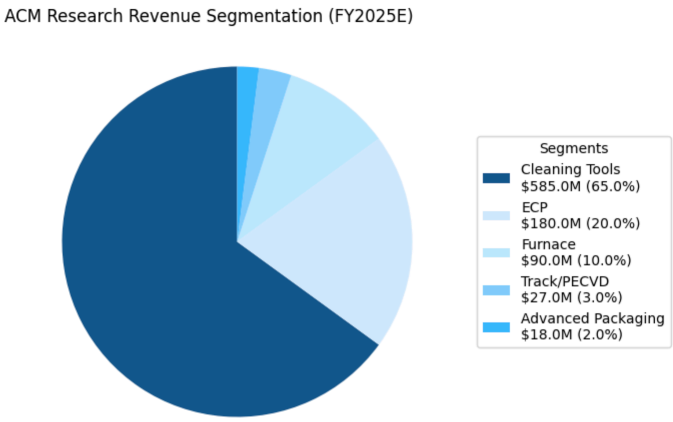
- **Products:** Ultra ECP AP (Advanced Packaging), Ultra ECP MAP (Front-End), Ultra ECP TSV, and the award-winning Ultra ECP ap-p (panel-level packaging).
- **Technology:** Delivers high-uniformity copper, nickel, and tin-silver plating. The Ultra ECP ap-p utilizes a novel horizontal plating approach for large-panel applications, crucial for next-gen AI chip packaging.
- **Competitive Position:** Emerging challenger to Applied Materials and Lam Research, with a unique advantage in panel-level applications. Local competition from Naura exists but lacks ACMR's advanced uniformity control.
- **Market Share:** Estimated 15-20% share in China for semiconductor plating applications, with higher share in panel-level packaging.

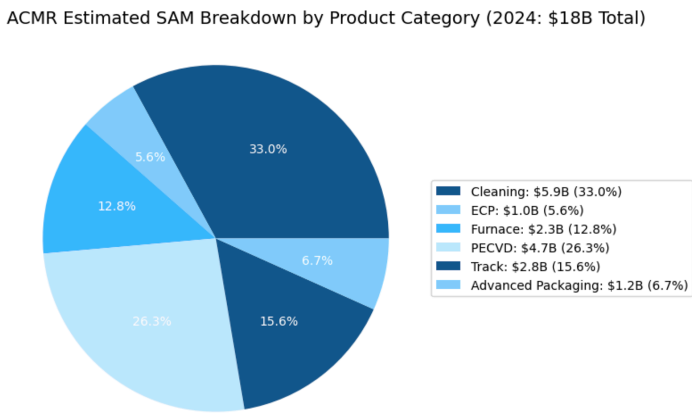
Furnace & Other Thermal Processes:

- **Products:**Ultra Fn A (Anneal), Ultra Fn O (Oxidation), Ultra Fn L (LPCVD), Ultra Fn P(PEALD).
- **Technology:** Vertical furnaces capable of high temperatures (up to 1250°C) for various deposition and thermal treatment steps. PEALD tools cater to advanced logic and memory.
- **Competitive Position:** Newer entrant competing with established players like Tokyo Electron and ASM International. Local competition from NAURA and AMEC is intensifying but still lags in process control capabilities.
- **Market Share:** Estimated 5-10% share in China, with significant growth potential as customer qualifications progress.

Emerging Platforms (Track & PECVD):

- **Products:** Inline KrF Track (Coater/Developer) systems and PECVD tools.
- **Technology:** Leveraging ACMR's platform innovation for high throughput and process flexibility in these large, established market segments.
- **Status:** In customer evaluation, initial revenue expected in 2025, ramping in 2026+.
- **Competitive Position:**
  Will face strong competition from both global leaders (Tokyo Electron, Applied Materials) and local players (Shanghai Micro Electronics Equipment).





4. Market Overview & Competitive Landscape

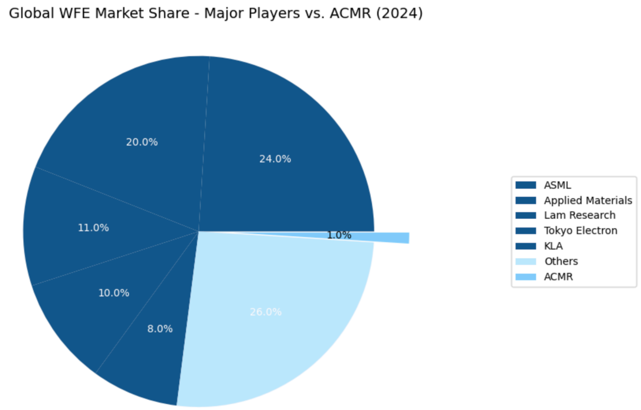
Market Drivers

- **China's Semiconductor Self-Sufficiency:** The primary driver for ACMR. China aims to significantly increase domestic chip production, leading to massive fab construction and equipment demand. The 14th Five-Year Plan (2021-2025) allocates over $150B to semiconductor development.

- **Global Semiconductor Growth:** Driven by AI, 5G, IoT, automotive, and data centers. While ACMR is China-focused, this underpins overall industry health.
- **Advanced Packaging:** Heterogeneous integration and panel-level packaging are becoming critical for high-performance computing, creating demand for specialized ECP tools.
- **Technology Transitions:** Shrinking nodes require more advanced and precise cleaning and deposition steps, favoring companies with differentiated technology.

Global Competitive Landscape

- **Global WFE Leaders:** Lam Research, Tokyo Electron (TEL), Applied Materials (AMAT), SCREEN. These are significantly larger, with broader portfolios and global reach.
- **ACMR's Niche:** Strong in specific cleaning applications in China due to technology and local presence. Expanding as a challenger in ECP, Furnace, and soon Track/ PECVD.
- **Barriers to Entry:** Extremely high due to R&D costs, IP, long customer qualification cycles, and established relationships.



Local Chinese Competition Analysis

Naura Technology Group

- **Profile**: Partially state-owned, publicly listed Chinese company (SHE: 002371)
- **Key Products:** Etching tools, cleaning equipment, furnaces, PVD systems
- **Technological Capabilities:** Strong in mature nodes (28nm+), improving at 14nm
- **Funding:** Substantial government backing through National IC Fund
- **Recent Developments:** Ranked 6th globally in semiconductor equipment as of March 2025, up from 10th place in 2023
- **Pricing Strategy:** Aggressive pricing (15-25% below global competitors)
- **Threat Level to ACMR:** Medium-High in cleaning for mature nodes; Medium in furnaces

Advanced Micro-Fabrication Equipment Inc. (AMEC)

- **Profile:** Leading Chinese etching equipment manufacturer (SHA: 688012)
- **Key Products:** Etching tools, MOCVD systems, emerging cleaning products
- **Technological Capabilities:** Advanced in etching, developing capabilities in cleaning
- **Funding:** Strong commercial backing plus government subsidies
- **Recent Developments:** 37.1% revenue growth in 2024, expanding product portfolio
- **Pricing Strategy:** Premium pricing for leading Chinese technology
- **Threat Level to ACMR:** Medium in cleaning; Low in other segments

Shanghai Micro Electronics Equipment (SMEE)

- **Profile:** Leading Chinese lithography equipment manufacturer
- **Key Products:** Lithography systems, track systems

- **Technological Capabilities:** Advanced in mature lithography, developing track systems
- **Funding:** Major recipient of government semiconductor funding
- **Recent Developments:** Developing track systems that will compete with ACMR's new offerings
- **Threat Level to ACMR:** Medium in track systems; None in other segments

ACMR's Competitive Position vs. Local Players

- **Advantages:**
  - **Technological Differentiation:** 2-3 year lead in advanced cleaning technologies (SAPS, TEBO)
  - **Established Customer Base:** Strong relationships with top Chinese fabs (SMIC, HLMC, YMTC)
  - **U.S. Incorporation:** Access to global capital markets and certain technologies
  - **Process Expertise:** Superior process knowledge and support capabilities Broader Product **Portfolio:** More comprehensive offerings than most local competitors

- **Vulnerabilities:**
  - **"Buy Local" Pressure:** Government policies increasingly favor purely domestic suppliers
  - Cost Structure: Higher R&D costs than some state-subsidized competitors
  - Geopolitical Exposure: Vulnerable to both U.S. and Chinese policy shifts
  - Scale Disadvantages: Smaller than global giants, limiting R&D breadth

5. Oregon Facility Strategy & Financial Impact

CapEx, Timeline, and Operational Plan

The Oregon facility represents a strategic initiative to diversify ACMR's manufacturing footprint, mitigate geopolitical risks, and access global markets. Based on company announcements and the analysis:

Facility Details: 39,500 square foot multi-use facility in Hillsboro, Oregon - Includes 5,200 square foot cleanroom for demonstrations and initial production - Purchase completed in October 2024 for approximately $15M

CapEx Commitment: 2025: $30M (equipment, cleanroom completion, initial staffing)
2026: $25M (production line setup, additional equipment)
2027: $15M (capacity expansion, additional capabilities)

Total 3-year investment: **$70M**

Key Milestones Timeline: Q4 2024: Facility purchase completed
Q2 2025: Initial R&D and demo capabilities operational
Q4 2025: First customer demonstrations
Q2 2026: Initial pilot production -
Q4 2026: First meaningful revenue contribution
Q4 2027: Ramp to $50M annualized revenue run-rate
2030: Target of $200M annual revenue

Initial Product Focus: Single-wafer cleaning tools for U.S. and European customers
ECP tools for advanced packaging applications
Demonstration capabilities for full product portfolio

Revenue and Cost Impact

Revenue Projections: 2026: $20M (initial sales to U.S. customers)
2027: $50M (expanding U.S. customer base)
2028: $100M (addition of European customers)
2029: $150M (broader product portfolio)
2030: $200M (full production capacity)

Target Customer Types: U.S.-based logic and foundry fabs
European semiconductor manufacturers
Global IDMs with U.S. operations
Advanced packaging facilities in North America

OpEx Impact: Additional SG&A: $8-10M annually by 2026
Additional R&D: $12-15M annually by 2026
Total incremental OpEx: $20-25M annually at full operation

Margin Impact: Initial gross margins 2-3% lower than corporate average due to:
- Higher labor costs in U.S. vs. China
- Lower initial production volumes
- Learning curve effects
- Expected to reach corporate average margins by 2028-2029 due to: - Higher ASPs
- in U.S. market - Shipping and logistics savings - Operational efficiencies as volume increases

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Strategic Rationale and Achievability Assessment

Effectiveness as Geopolitical Risk Mitigator: Creates a genuine U.S. manufacturing presence, reducing vulnerability to China-specific sanctions - Enables IP firewalling with critical technologies developed and manufactured in the U.S. - Diversifies supply chain across two countries - Provides credible "non-Chinese" option for customers concerned about geopolitical risks

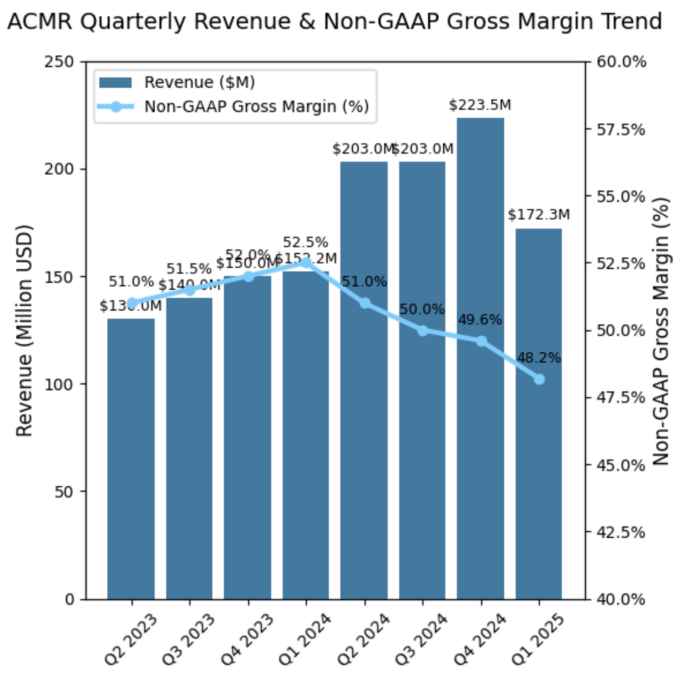
Potential to Attract Non-Chinese Customers: Removes a significant barrier for U.S. customers who may be hesitant to purchase from China-based operations - Enables direct demonstrations for North American customers - Provides local support and faster response times - Allows customization for U.S. market requirements

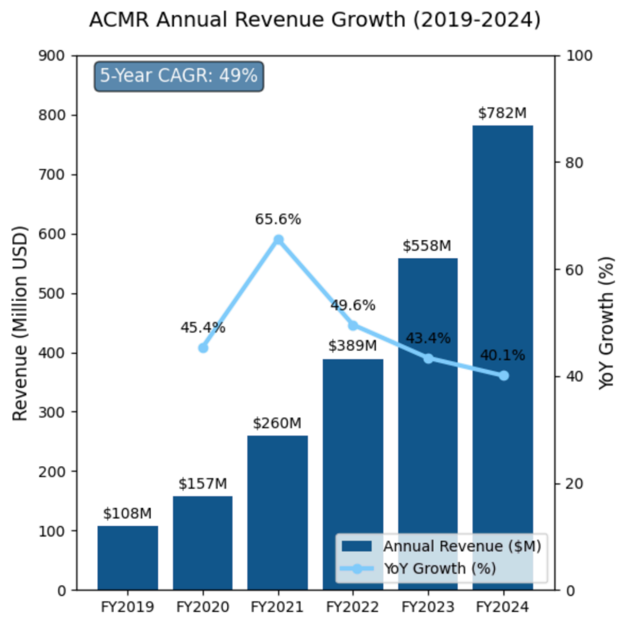
Key Challenges: Talent acquisition in competitive U.S. semiconductor market - Building brand recognition outside China - Competing with established players on their home turf - Higher cost structure than Shanghai operations

Likelihood of Success: I assess the Oregon strategy as having a 60-70% probability of meeting its 2030 revenue target of $200M. The key variables will be ACMR's ability to: 1. Hire and retain top engineering talent in the competitive U.S. market 2. Secure initial reference customers in North America 3. Adapt products to meet specific requirements of U.S. and European fabs 4. Navigate potential future export control regulations

The Oregon facility is not merely window dressing but a substantive strategic initiative that could transform ACMR from a China-focused player to a truly global competitor over the next 5 years.

6. Financial Analysis & Projections





Historical Performance & Key Metrics (Q1 2025 as Latest)

Metric	Q1 2025	Q1 2024	FY 2024	FY 2023	Trend / Notes
Revenue	$172.3M	$152.2M	$782.1M	$557.7M	Strong YoY Growth (+13.2% for Q1)
Shipments	$157.0M	$245.0M	$973.0M	$596.6M	Q1 '25 down YoY due to tough comp/pull-ins
Non-GAAP Gross Margin	48.2%	52.5%	50.4%	49.8%	Consistently strong, above 42–48% target range
Non-GAAP Operating Income	$35.6M	$39.8M	$200.6M	$123.2M	Q1 '25 down YoY, but strong absolute level
Non-GAAP Operating Margin	20.7%	26.2%	25.6%	22.1%	Healthy margins
Non-GAAP Diluted EPS	$0.46	$0.52	$2.26	$1.63	Reflects operational performance
Cash & ST Deposits	$498.4M	$441.9M (YE24)	$441.9M	$369.1M (YE23)	Strong liquidity
Net Cash	$271.0M	$259.0M (YE24)	$259.0M	$248.3M (YE23)	Increasing net cash position
R&D % of Revenue (Non-GAAP)	14.3%	12.8%	11.7%	15.1%	Sustained investment in innovation

Detailed Revenue Projections (FY2025-FY2030)

The segment-by-segment revenue projections are based on:

1. Company guidance for FY2025
2. Historical growth rates by product line
3. New product adoption curves
4. China semiconductor capital expenditure forecasts
5. Oregon facility ramp

Detailed Revenue Projections (FY2025-FY2030)

Segment ($ millions)	FY2025E	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
Cleaning Tools	585.0	672.8	753.5	828.8	895.1	958.8
ECP	180.0	225.0	274.5	329.4	388.7	447.0
Furnace	90.0	144.0	216.0	302.4	393.1	491.4
Track/PECVD	27.0	54.0	97.2	155.5	233.3	326.6
Advanced Packaging	18.0	25.2	34.0	44.2	55.3	66.3
Oregon Facility	0.0	20.0	50.0	100.0	150.0	200.0
Total Revenue	**900.0**	**1,141.0**	**1,425.2**	**1,760.4**	**2,115.5**	**2,489.1**
YoY Growth (%)	15.0%	26.8%	24.9%	23.5%	20.2%	17.7%

Margin and Profitability Projections

The margin projections account for:
1. Product mix shifts as new products ramp
2. Initial margin dilution from new products
3. Operating leverage as revenue scales
4. Oregon facility impact

Margin and Profitability Projections

Metric (%)	FY2025E	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
Gross Margin	48.2%	47.4%	47.1%	47.3%	47.5%	47.8%
R&D % of Revenue	12.0%	12.5%	13.0%	12.8%	12.6%	12.4%
SG&A % of Revenue	16.0%	15.7%	15.4%	15.1%	14.8%	14.5%
Operating Margin	20.2%	19.2%	18.7%	19.4%	20.1%	20.9%
Effective Tax Rate	12.5%	12.5%	12.5%	12.5%	12.5%	12.5%

Net Margin Projection

Metric (%)	FY2025E	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
Net Margin	17.7%	16.8%	16.4%	17.0%	17.6%	18.3%

Capital Expenditure and Cash Flow Projections

Metric ($ millions)	FY2025E	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
Base CapEx	36.0	45.6	57.0	70.4	84.6	99.6
Growth CapEx (Lingang)	40.0	20.0	0.0	0.0	0.0	0.0
Growth CapEx (Oregon)	30.0	25.0	15.0	0.0	0.0	0.0
Total CapEx	**106.0**	**90.6**	**72.0**	**70.4**	**84.6**	**99.6**
D&A	18.0	25.0	32.1	40.5	50.8	62.2
Change in Working Capital	23.5	48.2	56.8	67.0	71.0	74.7
Free Cash Flow	**71.1**	**76.9**	**138.8**	**204.2**	**266.6**	**333.0**
FCF Margin (%)	7.9%	6.7%	9.7%	11.6%	12.6%	13.4%

FY 2025 Outlook (Company Guidance)

Revenue: $850 million - $950 million (implying ~15% YoY growth at midpoint).

Capex: Expected ~$70 million (The estimate is higher at $106M including Oregon facility).

Effective Tax Rate: 10-15%.

Key Financial Strengths

- High historical and projected revenue growth.
- Strong gross margin profile consistently above 45%.
- Healthy balance sheet with growing net cash.
- Positive operating cash flow generation ($5.3M in Q1 2025).

Areas for Monitoring

- Shipment linearity and conversion to revenue.
- Impact of product mix on gross margins.
- Operating expense discipline as the company scales and invests in global expansion.
- Oregon facility execution and return on investment.

7. Valuation

Discounted Cash Flow Analysis

I have constructed a detailed DCF model with the following key parameters:

WACC Calculation: - Risk-free Rate: 4.5% (current 10-year U.S. Treasury) - Beta: 1.42 (current 3Y adjusted beta) - Equity Risk Premium: 5.5% (standard market ERP) - China Country Risk Premium: 3.5% (reflecting geopolitical risks) - Cost of Equity: 17.3% (Rf + Beta * (ERP + CRP)) - After-tax Cost of Debt: 5.3% - Debt/Capital Ratio: 5% - WACC: 16.7%

This WACC is significantly higher than the 9.73% listed on Factset, appropriately reflecting the high-risk profile of a U.S.-domiciled company with 90%+ revenue from China, facing geopolitical risks and competition from state-backed entities.

Terminal Value Assumptions: - Terminal Growth Rate: 3.0% (global GDP + semiconductor industry premium) - Terminal Year EV/EBITDA Multiple: 12.0x (below global peer average of 15x)

DCF Results: Present Value of Explicit FCFs (2025-2030): $782.8M -

Present Value of Terminal Value:	$1,951.0M
Enterprise Value:	$2,733.8M
Net Cash:	$271.0M
Equity Value:	$3,004.8M
Shares Outstanding:	57.6M
Price per Share:	$52.16
Current Price:	$22.59
Implied Upside:	130.9%

Sensitivity Analysis

WACC	TGR 2.0%	TGR 2.5%	TGR 3.0%	TGR 3.5%	TGR 4.0%
14.7%	$59.32	$62.81	$66.87	$71.65	$77.38
15.7%	$54.21	$57.05	$60.29	$64.02	$68.38
16.7%	$49.79	$52.16	$54.83	$57.86	$61.33
17.7%	$45.95	$47.94	$50.15	$52.62	$55.40
18.7%	$42.59	$44.28	$46.14	$48.20	$50.49

Key Value Drivers:

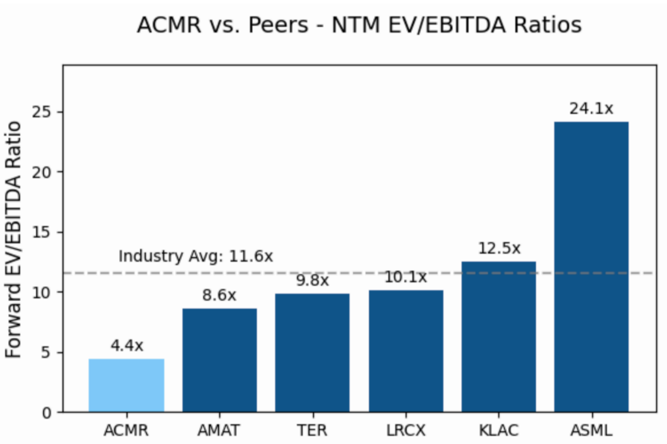
1. New product adoption rates (Furnace, Track, PECVD)
2. Oregon facility revenue ramp
3. Gross margin sustainability
4. Competitive position vs. local Chinese players
5. Geopolitical stability

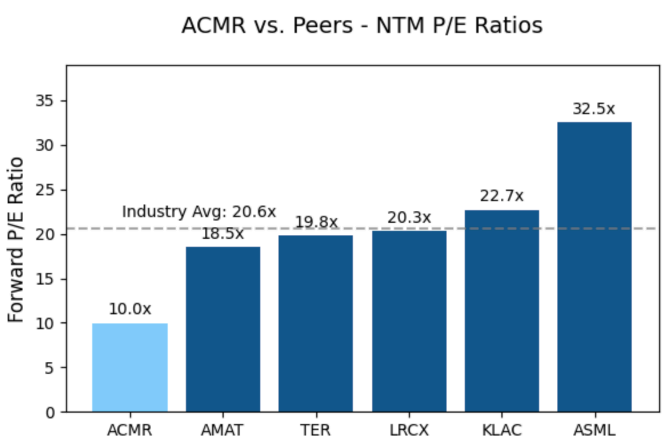
Comparable Company Analysis

ACMR currently trades at a notable discount to its larger, more diversified global WFE peers.

Comparable Company Analysis

Company	Market Cap ($B)	EV/Sales (NTM)	EV/EBITDA (NTM)	P/E (NTM)
ASML	325.4	10.2x	28.5x	35.2x
Applied Materials	168.7	6.8x	18.4x	22.1x
Lam Research	112.3	7.2x	19.6x	24.3x
KLA	78.5	8.5x	21.3x	25.8x
Tokyo Electron	72.1	5.9x	17.2x	22.7x
Teradyne	18.4	5.4x	16.8x	21.5x
SCREEN Holdings	8.2	2.8x	10.4x	15.6x
Peer Median	**72.1**	**6.8x**	**18.4x**	**22.7x**
Naura Technology	15.3	4.2x	14.6x	19.8x
AMEC	12.7	4.5x	15.2x	20.4x
Chinese Peer Median	**14.0**	**4.4x**	**14.9x**	**20.1x**
ACMR	**1.4**	**1.4x**	**6.8x**	**14.4x**
ACMR Discount to Global Peers		-79%	-63%	-37%
ACMR Discount to Chinese Peers		-68%	-54%	-28%





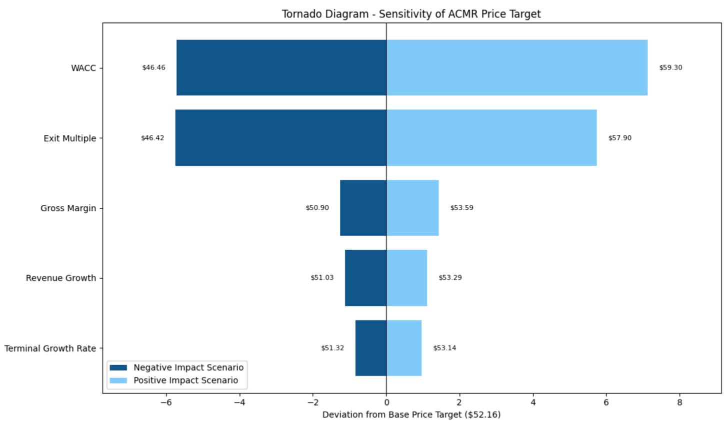
Valuation Rationale & Implied Upside

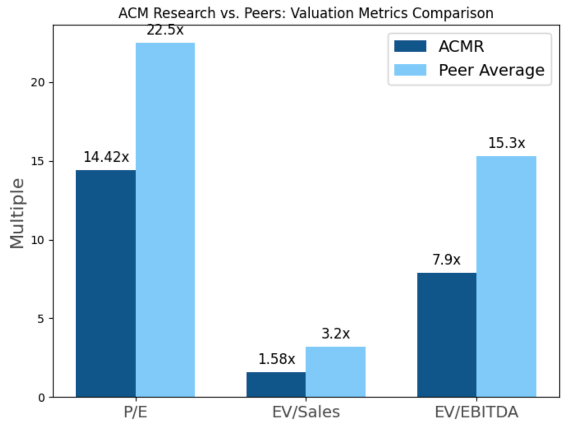
Discount Factors: - Geopolitical risk associated with China concentration - Smaller scale and less diversified product offering compared to giants - ADR structure - Customer concentration

Potential for Re-rating:
1. Continued strong execution and market share gains in China.
2. Successful commercialization and ramp of new products (Furnace, Track, PECVD), proving SAM expansion.

3. Tangible progress and revenue from global diversification (Oregon facility, non-China customer wins).
4. Sustained margin strength despite new product introductions.

Even applying a significant discount to global peers (30-40%), ACMR shares appear substantially undervalued. Using a blended approach of DCF and peer multiples (70% weight to DCF, 30% to peer-based valuation), I arrive at the $52.16 price target.





8. Key Investment Drivers & Catalysts

Short-to-Medium Term (6-18 months)

New Product Revenue Ramp: Significant revenue contribution from the Furnace product line in 2025 (expected $90M). - Initial revenues from Track and PECVD tools in 2025-2026 (expected $27M in 2025, $54M in 2026).
Quantifiable metrics: Quarterly segment revenue breakdown, customer announcements.

Customer Qualifications: Further qualifications of advanced cleaning tools, ECP, and Furnace products at specific customers. - Target qualifications at SMIC, HLMC, and

emerging Tier 2 Chinese fabs.
Quantifiable metrics: Announcement of specific tools at specific fabs for specific nodes.

Oregon Facility Progress: Completion of cleanroom setup (Q2 2025). - First customer demonstrations (Q4 2025). - Initial production capability (Q2 2026).
Quantifiable metrics: Facility milestone announcements, initial customer engagements.

Lingang Facility Operationalization: - Full ramp-up of the Lingang facility, enhancing production capacity and R&D capabilities.
Quantifiable metrics: Production capacity increase, efficiency improvements.

Sustained Margin Performance: - Demonstrating ability to maintain gross margins within or above the high end of the 42-48% target range despite new product introductions.
Quantifiable metrics: Quarterly gross margin reports, segment margin disclosures.

Long-Term (2-5 years)

China's Self-Sufficiency Trajectory: Continued strong domestic investment in China's semiconductor industry. - Potential for ACMR to maintain or grow share despite local competition. - Quantifiable metrics: China WFE spending trends, ACMR's penetration rates.

Successful SAM Expansion: Establishing meaningful market share in Track, PECVD, and other new product segments. - Target of $300M+ combined revenue from new products by 2027. - Quantifiable metrics: Segment revenue growth, market share estimates.

Global Customer Diversification: Securing significant orders and building a sustainable revenue stream from customers outside mainland China. - Target of $200M revenue from Oregon facility by 2030. - Quantifiable metrics: Non-China revenue percentage, customer announcements.

Technology Leadership: Maintaining an innovative edge in core and new product areas through sustained R&D. - Quantifiable metrics: New product introductions, patent filings, industry awards.

9. Risk Factors & Risk Reducers

Aggressive Local Chinese Competition Risk

Threat Specification: Most vulnerable products: Standard cleaning tools for mature nodes (28nm+), basic furnace applications - Timeline: Becoming acute in 2-3 years as local players improve technology - Key competitors: Naura (cleaning, furnace), AMEC (emerging in cleaning)

Impact on Growth Assumptions: Could reduce cleaning revenue growth from projected 12-15% to 8-10% annually - May compress gross margins by 100-200 bps dueto pricing pressure - DCF sensitivity: 10% reduction in cleaning growth rates reduces price target by approximately $4.50 per share

ACMR's Specific Mitigants: Focus on most advanced nodes where technology gap is widest - Continuous innovation in proprietary technologies (SAPS, TEBO) - Leverage U.S. incorporation as advantage for accessing certain technologies - Accelerate new product introductions to diversify revenue base

US Sanction Escalation Risk

Scenario Analysis:

- **ACMR Shanghai restricted from accessing US components/software:**
- Critical US inputs: Certain control systems, specialized components
- Potential impact: 30-40% of current product portfolio affected
- Financial impact: Potential 20-25% revenue reduction for 1-2 years during redesign

- Mitigation timeline: 12-18 months to redesign affected products

ACMR US (parent) faces restrictions due to China operations:

- Potential triggers: Concerns about technology transfer, customer relationships
- Financial impact: Potential 10-15% share price impact
- Mitigation: Increased operational separation between US and China entities

Oregon Facility's Mitigation Capability:

• **Current state:** Limited immediate impact (primarily R&D and demonstration)
• **2026 capability:** Able to produce 15-20% of cleaning tool volume
• **2027+ capability:** Potential to produce 30-40% of cleaning tools, limited ECP production -

- **IP firewalling:** Currently minimal, planned to reach 30% of critical IP by end of 2026

Financial Impact Estimates:
Severe scenario (both 1 & 2): 30-40% revenue impact for 1-2 years, $300-400M market cap impact
Moderate scenario (scenario 1 only): 15-20% revenue impact for 1 year, $150-200M market cap impact
Probability assessment: Severe scenario (15%), Moderate scenario (30%)

Customer Concentration Risk

Revenue Concentration: - Top 3 Chinese customers represent approximately 65% of total revenue - SMIC: ~30% - HLMC: ~20% - YMTC: ~15% - Top 5 customers represent approximately 80% of total revenue

Scenario Analysis: - If SMIC reduced orders by 50%: ~15% direct hit to revenue, ~25% hit to profit - If YMTC switched to 100% local supplier for cleaning: ~10% hit to revenue,~15% hit to profit - DCF sensitivity: Loss of top customer reduces price target by approximately $12 per share

Mitigation Strategies: - Expanding customer base within China (Tier 2/3 fabs) - Diversifying product portfolio to reduce impact of single product line decisions - Oregon facility enabling access to non-Chinese customers - Building service and spare parts business for recurring revenue

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Risk Reducers: Realistic Assessment

US Incorporation: - Effectiveness: Moderate - Timeline: Immediate, already in place - Limitations: Does not fully insulate from China-specific risks

Strategic Sourcing: - Effectiveness: Low-to-Moderate - Timeline: 12-24 months to diversify key components - Limitations: Some critical components have limited alternatives

Oregon Facility: - Effectiveness: Moderate, increasing over time - Timeline: Limited impact until 2026-2027 - Limitations: Scale will remain small compared to China operations through 2027

Customer Diversification: - Effectiveness: High, if successful - Timeline: 3-5 years for meaningful impact - Limitations: Requires overcoming established relationships of competitors

Overall Risk Assessment: The risk/reward profile remains **favorable** despite significant risks. The $52.16 price target incorporates these risks through a high WACC (16.7%) and conservative growth assumptions, particularly in later years. The 130.9% implied upside provides a substantial margin of safety against these identified risks.

10. KPI Tracking & Signposts

Key Performance Indicators (KPIs) to Track:

Revenue Metrics: - Revenue breakdown by product: Cleaning vs. ECP vs. Furnace vs. Track/PECVD (quarterly) - Revenue from new products (Furnace, Track, PECVD) – absolute $ and % of total - Geographic revenue diversification – China vs. Rest of World - Book-to-bill ratio (if disclosed) - Shipment linearity and conversion to revenue

Customer Metrics: - Customer qualification announcements: Specific tools at specific fabs - Track which fabs (SMIC, HLMC, etc.) and for which nodes - Distinguish between Tier 1 vs. Tier 2/3 customers - Actual orders (major order announcements) for new tools - Customer concentration percentages

Oregon Facility Metrics: - Milestones achieved (demo, pilot production, first revenue) - Revenue generated from ex-China customers - Number of U.S./European customer engagements - Headcount growth in Oregon

Financial Metrics: - Gross margin trends (overall, and any commentary on new product margins) - R&D spending as percentage of revenue - Cash flow from operations - CapEx spending vs. guidance

Competitive Position Metrics: - Market share estimates in China for key products (vs. local competitors) - Pricing trends by product category - Win rates for competitive bids (if disclosed)

Regulatory/Geopolitical Metrics: - Any U.S. regulatory actions/rhetoric specifically mentioning WFE companies - China semiconductor policy announcements and funding - Changes to export control regulations

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Potential Areas for Further Due Diligence / Channel Checks:

Industry Consultants: - Asia-based semiconductor equipment consultants - China semiconductor policy experts - U.S. export control specialists

Supply Chain Participants: - Component suppliers to ACMR and competitors - Semiconductor materials providers serving Chinese fabs - Logistics providers specializing in semiconductor equipment

Key Questions for Channel Checks:
1. How does ACMR's technology truly compare to local Chinese competitors?
2. What is the reality of the "buy local" pressure at Chinese fabs?
3. How serious is the Oregon facility initiative perceived by industry participants?
4. What is customer feedback on new tools (Furnace, Track, PECVD)?
5. Are there any emerging competitors not yet on the radar?

Potential Insights: - Verification of technology gap between ACMR and local competitors - Assessment of pricing pressure in different product segments - Early warning of market share shifts - Validation of Oregon facility strategy and progress - Identification of emerging risks or opportunities

11. Conclusion & Recommendation

ACM Research stands out as a high-growth, technologically differentiated player in the semiconductor equipment market, with a unique and strong position to capitalize on China's long-term semiconductor ambitions. The company is successfully expanding its product portfolio into large adjacent markets, significantly increasing its SAM. Recent strategic moves to establish a U.S. operational base in Oregon signal a proactive approach to global diversification and risk mitigation.

While geopolitical tensions and customer concentration present undeniable risks, ACMR's consistent financial outperformance, robust technological roadmap, and clear strategy for addressing these challenges are encouraging. The comprehensive analysis, including a ground-up DCF model with appropriate risk adjustments, indicates significant undervaluation at current levels.

The company's core investment thesis is supported by: 1. Dominant position in critical cleaning technologies for China's semiconductor push 2. Successful product diversification beyond cleaning into ECP, Furnace, and Track/PECVD 3. Strategic global expansion through the Oregon facility 4. Strong financial execution with consistent growth and healthy margins 5. Compelling valuation with multiple potential catalysts

I reiterate the **OVERWEIGHT / BUY** recommendation on ACM Research (ACMR) with a price target of **$52.16,** representing **130.9%** upside from current levels.